

6-MONTH MASTER STUDY TIMETABLE: AI/ML & QUANTITATIVE TRADING

Timeline: October 5, 2026 – April 4, 2027 Daily Window: 7:00 PM – 11:00 PM (4.0 Hours) Allocation: 70% AI/ML | 30% Systematic Trading

1. Operational Evening Timeboxing Architecture			Monday – Sunday Schedule Profiles
DAY TYPE	PRIMARY BLOCK (AI/ML)	SECONDARY BLOCK (TRADING)	TARGET FOCUS & BREAKS
Monday – Friday	19:00–20:30: Core Theory 20:45–22:00: Coding Lab	22:10–23:00: Market Mechanics, Structure & Rules	Two 10–15 min screen-off cognitive breaks. Daily concrete code commit and journal entry.
Saturday	19:00–21:00: Pipeline Engineering & Project Build	21:15–23:00: Manual & Vectorized Strategy Backtesting	Deep integration block. No new theory; strictly hands-on builds and backtesting logs.
Sunday	19:00–20:30: Theory Review & Documentation	20:45–21:45: Sector Relative Strength & Journal Audit	21:45–23:00: 40-Point Weekly Retrospective Audit and upcoming week setup.

2. Six-Month Macro Progression & Milestones Matrix				October 2026 → April 2027
PHASE	PRIMARY TRACK (AI/ML)	SECONDARY TRACK (TRADING)	MAJOR MONTHLY MILESTONE	EVALUATION GATE
Month 1 Oct 05–Nov 01	NumPy vectorization, Pandas wrangling, stats, CLT, Matplotlib/Seaborn.	Indian market structure, order book, candlesticks, S/R, volume, ATR.	Automated EDA Diagnostic Engine: Standalone CLI data profiling script.	Broadcast matrix Euclidean distance without loops; mark 10 clean S/R flips.
Month 2 Nov 02–Nov 29	Regression, Gradient Descent, Logistic, Trees, Ensembles, XGBoost.	Moving averages, RSI, ATR stops, 1% risk rule, trade journal, statements.	Enterprise Default Predictor: Leak-free XGBoost classification pipeline.	Derive normal equation; log 60 manual backtested trades with 1% risk.
Month 3 Nov 30–Jan 03	Perceptrons, backprop math, PyTorch autograd, optimizers, CNNs.	Backtesting biases (lookahead, survivorship), expectancy, Sharpe, MDD.	PyTorch Vision Classification System: Custom CNN with TensorBoard.	Hand-derive backprop; deliver 100-trade empirical audit dossier.
Month 4 Jan 04–Jan 31	Sequence models, Multi-Head Attention, Transformers, embeddings.	Vectorized backtesting in Pandas, walk-forward testing, valuation ratios.	Mini-Transformer & Quant Engine: Scratch attention block + backtest suite.	Derive attention complexity; verify Pandas backtest with t+1 shift.
Month 5 Feb 01–Feb 28	Production RAG, hybrid search, agents, FastAPI, Docker Compose.	Derivatives theory, futures margins, options mechanics, Greeks, IV skew.	Enterprise Knowledge Assistant: Containerized Hybrid RAG REST service.	Deploy local containerized API; prove unhedged OTM options negative edge.
Month 6 Mar 01–Apr 04	Full-stack multi-agent platform, CI/CD, cloud deployment, monitoring.	Forward live paper trading (₹0 capital), slippage tracking, discipline log.	UNIFIED CAPSTONE: FinSpectra Financial Intelligence Platform.	Live defense of cloud platform + 30-day paper-trading audit.

CORE OPERATING RULES OF ENGAGEMENT Prioritization: AI/ML is primary. Trading reduces to zero if college exam loads demand it. Zero-Real-Money Mandate: Exactly ₹0 deployed across all 6 months. Absolute paper trading. Anti-Gambling Directive: Zero telegram tips, zero intraday scalping, zero unhedged leverage.	WEEKLY EVALUATION METRIC (SUNDAY 21:45) AI/ML Theory & Code Implementation: 20 Points (Target: ≥ 17/20) Trading Mechanics & Execution Discipline: 10 Points (Target: ≥ 9/10) Consistency & Timetable Adherence: 10 Points (Target: ≥ 8/10) Threshold: If score < 34/40, reduce trading hours the following week.
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MONTH 1: DAILY STUDY SCHEDULE (WEEKS 1 & 2)

Phase Focus: Vectorized Computation, Linear Algebra & Indian Market Microstructure

Week 1: NumPy Array Internals & Exchange Architecture					October 05 – October 11, 2026
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Oct 05	Mon	Conda envs, Git repo setup, NumPy ndarray memory strides, contiguous blocks.	Indian market structure: SEBI, RBI roles; primary (IPO) vs secondary markets.	Configure repo; verify BLAS/LAPACK acceleration in NumPy installation.	Pushed GitHub repo with verified environment script.
Oct 06	Tue	Array generation (arange, linspace), reshaping, flattening, view vs deep copy.	NSE vs BSE; pre-open (9:00–9:15), regular trading; NIFTY 50 and SENSEX.	Execute 10 memory-efficient array operations; test np.shares_memory().	Jupyter notebook logging view vs copy stride mutations.
Oct 07	Wed	Broadcasting rules, axis dimensions, profiling loops vs vectorized ops (%timeit).	Demat vs Trading accounts, clearing corps (NSCCL), T+1 settlement cycles.	Benchmark pure loop vs vectorized array math over 10M elements.	Benchmark script showing >50x vectorized speedup.
Oct 08	Thu	Boolean masking, fancy indexing, condition-based extraction (np.where, np.select).	Order types: Market, Limit, SL-L, SL-M, AMO; liquidity impact of market orders.	Implement conditional thresholding on multi-dimensional sensor array.	Script isolating and replacing outliers using boolean masks.
Oct 09	Fri	Matrix operations: Dot product, matmul (@), transpose, identity, matrix inversion.	The Order Book: Bid, Ask, Spread, Depth of Market (DOM); matching engine.	Write matrix multiplication using nested loops; replicate with np.matmul.	Unit test verifying identical results & speed ratio.
Oct 10	Sat	Lab Build: Euclidean distance, Cosine similarity, Manhattan distance.	Chart Setup: TradingView setup, OHLC candlesticks anatomy and wicks.	Build vector_math.py library with automated test assertions.	Documented math module verified against Scipy.
Oct 11	Sun	NumPy revision, Git repo cleanup, week summary notes.	Inspect 10 daily candles on NIFTY 50; identify balance vs rejection wicks.	Compile Week 1 retrospective; score study consistency.	Weekly self-audit log and clean chart layout saved.
Week 2: Linear Algebra, Pandas Ingestion & Structural Price Action					October 12 – October 18, 2026
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Oct 12	Mon	Vectors in $\mathbb{R}^n$, linear independence, span, vector projection, basis vectors.	Price Action core premise; Auction market theory: balance vs imbalance.	Code vector projection in NumPy; project 2D vector onto arbitrary basis.	Matplotlib plot displaying projected basis vectors.
Oct 13	Tue	Systems of linear equations: Gaussian elimination, matrix rank, np.linalg.solve.	Market Structure: Swing Highs and Swing Lows. HH & HL vs LH & LL.	Solve inconsistent system via least-squares (np.linalg.lstsq).	Notebook solving exact vs approximate linear systems.
Oct 14	Wed	Singular Value Decomposition (SVD) and PCA geometric intuition.	Trend classification: Up, Down, and Sideways ranges across timeframes.	Implement image compression using truncated SVD on grayscale image.	Script rendering image at 5, 20, 50 singular values.
Oct 15	Thu	Pandas Series & DataFrames: Ingestion (CSV, Parquet), .loc vs .iloc indexing.	Support & Resistance (S/R): Horizontal zones vs lines; round numbers.	Load 100k-row dataset; execute complex multi-column index filtering.	Leak-free filtering notebook with zero chained index warnings.
Oct 16	Fri	Handling missing values (imputation, dropping), duplicates, type casting (astype).	S/R Polarity Shifts: Broken support turning into resistance and vice versa.	Write a robust data cleaner handling dirty strings, nulls, and types.	Automated cleaning function returning audit log.
Oct 17	Sat	Lab Build: Data ingestion and	Chart Lab: Mark 10	Run validator across 3	Reusable validator

MONTH 1: DAILY STUDY SCHEDULE (WEEKS 3 & 4)

Phase Focus: Advanced Wrangling, Inferential Statistics & Technical Foundations

Week 3: Split-Apply-Combine, Visual Pipelines & Pattern Context					October 19 – October 25, 2026
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Oct 19	Mon	Pandas groupby, multi-level aggregations, transform, and custom apply methods.	Volume analysis basics: Volume as confirmation; breakouts vs low-volume drift.	Calculate multi-level aggregations across sector categories.	Script computing group mean, variance, rank transforms.
Oct 20	Tue	Merging, joining (inner, outer, left, right), concatenation, schema handling.	Volume Price Analysis (VPA): Accumulation and distribution signatures.	Merge transaction tables with user metadata; handle orphan keys.	Merge script with automated row-count sanity checks.
Oct 21	Wed	Time-series Pandas: DatetimeIndex, resampling, shifting, rolling windows.	Candlesticks in context: Pin bars (Hammers/Shooting Stars), Engulfing bars.	Calculate 20-day rolling mean, rolling std, and daily returns on equity data.	Rolling statistics series and forward returns script.
Oct 22	Thu	Matplotlib object-oriented API (fig, ax), multi-panel subplots, styling.	Classical patterns: Double tops/bottoms, Head and Shoulders, Triangles.	Build publication-ready 4-panel figure illustrating distributions.	High-res 300 DPI multi-panel plot exported to disk.
Oct 23	Fri	Seaborn: Heatmaps, pairplots, violin plots, box plots, empirical CDFs.	Flags, Pennants, and Bull/Bear channels; continuation vs reversal setups.	Generate full correlation matrix heatmap with masked upper triangle.	Annotated correlation matrix plot with colorbar.
Oct 24	Sat	Lab Build: Ingest daily equity data using yfinance; build multi-asset dashboard.	Chart Lab: Document 5 Head & Shoulders or Double Bottoms on historical charts.	Normalize 5 stocks to base 100; plot drawdown curves.	Drawdown visualizer script + 5 annotated charts.
Oct 25	Sun	Review Pandas grouping and Matplotlib figure layout routines.	Scan NIFTY sectoral indices (IT, Bank, Auto) for relative strength.	Synthesize market structure summary; complete weekly review.	Sector relative-strength comparison table.
Week 4: Probability Distributions, Hypothesis Testing & Dynamic ATR Stops					October 26 – November 01, 2026
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Oct 26	Mon	Descriptive stats: Central tendency, dispersion (variance, std, IQR), skew/kurtosis.	Moving Averages: SMA vs EMA; mathematical weighting differences.	Write pure NumPy functions for skewness and kurtosis; verify with SciPy.	Analytical match verified against scipy.stats.
Oct 27	Tue	Probability: Random variables, PMF, PDF, CDF; discrete vs continuous distributions.	Moving Averages: 50 EMA and 200 EMA as dynamic support and resistance.	Simulate coin/dice trials via np.random; verify Law of Large Numbers.	Convergence plot illustrating theoretical distributions.
Oct 28	Wed	Normal distribution, standard normal, Z-scores, 68-95-99.7 empirical rule.	Relative Strength Index (RSI): Formula, overbought/oversold myths, 50 centerline.	Standardize raw dataset to Z-scores; flag points with $ Z > 3.0$.	Anomaly detector script filtering statistical outliers.
Oct 29	Thu	Central Limit Theorem (CLT): Sampling distributions, standard error of the mean.	RSI Divergences: Regular (reversal) vs Hidden (continuation) divergences.	Draw samples from exponential distribution; plot normal sample means.	4-panel graphic proving CLT convergence empirically.
Oct 30	Fri	Inferential stats: Hypothesis testing, null/alt hypothesis, p-values, t-test.	Average True Range (ATR): Volatility measurement; setting dynamic stops.	Execute two-sample t-test comparing feature means across partitions.	Statistical test report with p-value significance.
Oct 31	Sat	Capstone Build: Assemble Automated EDA Diagnostic	Chart Lab: Mark 10 charts showing 50/200	Package EDA code into CLI tool exporting	GitHub repo with eda_engine.py

MONTH 2: DAILY STUDY SCHEDULE (WEEKS 5 & 6)

Phase Focus: Supervised Regression, Gradient Descent, Classification & Dynamic Risk

Week 5: Supervised Regression, Gradient Descent & Risk Management 101					November 02 – November 08, 2026
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Nov 02	Mon	Supervised learning formulation: Features, targets, train/val/test splits, leakage.	Golden Rules of Trading: Capital preservation, probability mindset, anti-gambling.	Implement automated dataset splitter in NumPy; verify index isolation.	Leak-free partition function with unit test.
Nov 03	Tue	Simple Linear Regression: Hypothesis, Mean Squared Error (MSE) cost function.	Risk Management 101: Defining Capital at Risk (R); the strict 1% account rule.	Derive OLS normal equation: $\beta = (X^T X)^{-1} X^T y$ in pure NumPy.	Analytical script fitting linear model to synthetic data.
Nov 04	Wed	Gradient Descent: Batch, Mini-batch, Stochastic (SGD); learning rate tuning.	Risk-to-Reward Ratio (R:R): Minimum 1:2 parameters; expectancy curves.	Build Batch Gradient Descent from scratch; plot MSE cost curve.	Jupyter notebook showing monotonic cost reduction.
Nov 05	Thu	Multiple Linear Regression: Feature scaling, collinearity, interpreting weights.	Position Sizing Math: Calculating exact shares from stop distance & fixed risk.	Code multi-variable regression class with .fit() and .predict() methods.	Reusable Python class matching Scikit-learn API design.
Nov 06	Fri	Scikit-learn Linear Models: Residual analysis, homoscedasticity, R ² , MAE, RMSE.	Mechanical Trend Setup: Formulate EMA 20 pullback rules in 50/200 uptrend.	Fit linear model; plot residual distributions and Q-Q normality plots.	Diagnostic notebook checking regression assumptions.
Nov 07	Sat	Lab Build: End-to-end regression model; cross-validation across folds.	Chart Lab: Find 10 historical EMA 20 pullbacks on NIFTY stocks; log setups.	Write automated pipeline evaluating variance across k-fold splits.	Cross-validated regression script in repository.
Nov 08	Sun	Review gradient descent convergence mechanics and normal equation math.	Build watchlist of Indian stocks trading above rising 50 and 200 EMAs.	Weekly self-audit; verify position-sizing formula execution.	Watchlist of 10 strong trend candidates logged.
Week 6: Regularization, Logistic Classification & Dynamic ATR Stops					November 09 – November 15, 2026
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Nov 09	Mon	Regularization: Bias-variance tradeoff; Ridge Regression (L2 penalty) weight decay.	Stop-Loss Placement: Technical invalidation vs emotional exits; ATR stops.	Implement Ridge cost function in NumPy; track coefficient shrinkage.	Plot showing weight shrinkage across lambda values.
Nov 10	Tue	Lasso Regression (L1 penalty): Sparsity, feature pruning, L1 vs L2 geometry.	Profit Taking Mechanics: Fixed R:R targets vs trailing stops using swing lows.	Fit Lasso on collinear dataset; verify non-informative weights hit zero.	Script proving automated feature pruning via Lasso.
Nov 11	Wed	Logistic Regression: Odds, Logit, Sigmoid function, Binary Cross-Entropy loss.	The Trading Journal: Required fields (Date, Ticker, Setup, R, Result, Error).	Derive Sigmoid and binary log-loss in NumPy; prevent log(0) numerical errors.	Stable custom binary cross-entropy loss function.
Nov 12	Thu	Multi-class classification: Softmax, Categorical Cross-Entropy, One-vs-Rest (OvR).	Trading Psychology: FOMO, revenge trading, disposition effect.	Build two-class Logistic Regression from scratch using gradient updates.	Script plotting 2D linear decision boundary.
Nov 13	Fri	Classification Metrics: Confusion matrix, Precision, Recall, F1, ROC-AUC.	Strategy 1 Document: Mechanical Trend-Following rules and stop-loss logic.	Write custom functions computing Precision, Recall, F1 without sklearn.	Code file validated against sklearn.metrics.
Nov 14	Sat	Lab Build: Loan default predictor; tune decision	Chart Lab: Apply Strategy 1 to 20	Plot Precision-Recall curve; set operational	Threshold-tuning notebook + 20

MONTH 2: DAILY STUDY SCHEDULE (WEEKS 7 & 8)

Phase Focus: Decision Trees, Ensembles, XGBoost & Fundamental Statement Analysis

Week 7: Decision Trees, Random Forests & Financial Statement Analysis					November 16 – November 22, 2026
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Nov 16	Mon	Decision Trees: Splitting criteria, Information Gain, Shannon Entropy, Gini.	Strategy 2 Document: Mean Reversion concept; oversold pullbacks in uptrends.	Code Entropy and Gini Impurity from scratch for class distributions.	Script proving Gini and Entropy yield identical split points.
Nov 17	Tue	Tree constraints: max_depth, min_samples_split, min_samples_leaf; overfitting.	Mean Reversion rules: RSI < 35 while above 200 EMA; reversal candle entry.	Train DecisionTreeClassifier; visualize tree structure via plot_tree.	High-resolution diagram illustrating decision tree splits.
Nov 18	Wed	Ensembles: Wisdom of crowds, Bootstrap Aggregation (Bagging), OOB error.	Risk parameters for Mean Reversion: Tight ATR stops, rapid profit taking at 20 EMA.	Build bootstrap sampler in NumPy; train 10 unconstrained decision trees.	Script showing variance reduction via bagging.
Nov 19	Thu	Random Forests: Feature bagging, MDI importance vs permutation importance.	Fundamental Analysis 1: Income Statement, Balance Sheet, Cash Flow Statement.	Fit RandomForestClassifier; compare Gini importance with permutation metric.	Feature importance bar chart identifying drivers.
Nov 20	Fri	Boosting Foundations: AdaBoost mechanics, sample reweighting, sequential fitting.	Fundamental Analysis 2: Valuation ratios: P/E, P/B, EV/EBITDA, Debt-to-Equity.	Implement AdaBoost sample weight updates in pure Python.	Notebook showing exponential weight shifts on errors.
Nov 21	Sat	Lab Build: Credit scoring model comparing Trees, Random Forest, AdaBoost.	Chart Lab: Apply Strategy 2 to 20 historical trades on HDFCBANK/KOTAK.	Benchmark tree models across accuracy, latency, and ROC-AUC.	Comparative benchmark study + 20 trades logged.
Nov 22	Sun	Review tree splitting criteria and ensemble variance reduction mechanics.	Read quarterly financial reports of 2 NIFTY 50 companies; check cash flows.	Compile fundamental analysis notes; complete weekly review.	Fundamental balance sheet notes in repository.
Week 8: Gradient Boosting (XGBoost), Hyperparameter Tuning & Strategy Audit					November 23 – November 29, 2026
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Nov 23	Mon	Gradient Boosting Machines (GBM): Gradient descent in function space, residuals.	Fundamental Analysis 3: ROE and ROCE; economic moats and competitive advantages.	Code toy gradient booster iteratively fitting trees on residual errors.	Script illustrating residual error reduction across stages.
Nov 24	Tue	XGBoost Architecture: Histogram-based splitting, missing value handling.	Fundamental Analysis 4: Red flags: promoter pledging, auditor exits, debt spikes.	Train XGBClassifier; optimize parameters; profile memory and compute speed.	Notebook running XGBoost with early stopping.
Nov 25	Wed	Hyperparameter Optimization: Grid vs Random vs Bayesian Search (Optuna).	Trading Plan Integration: Pre-market preparation, execution rules, daily review.	Write Optuna study optimizing XGBoost hyperparameters over 50 trials.	Convergence plots and parameter importance charts.
Nov 26	Thu	Data Preprocessing: Scikit-learn Pipeline and ColumnTransformer abstractions.	Expectancy Formula: $E = (\text{Win}\% * \text{Avg Win}) - (\text{Loss}\% * \text{Avg Loss})$.	Build leak-free preprocessing pipeline handling categorical & numeric data.	Reusable Scikit-learn pipeline saved to joblib.
Nov 27	Fri	Unsupervised Learning: k-Means clustering, inertia, elbow method, silhouette score.	Consolidate Strategy 1 & Strategy 2 backtest trades (40 total); calculate E.	Implement k-Means from scratch in NumPy; iterate cluster centroids.	Custom k-Means class plotting centroid paths.

MONTH 3: DAILY STUDY SCHEDULE (WEEKS 9 & 10)

Phase Focus: Deep Learning Foundations, Autograd & Quantitative Strategy Backtesting

Week 9: Multi-Layer Perceptrons, Backpropagation & Backtest Pitfalls					November 30 – December 06, 2026
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Nov 30	Mon	Artificial Neurons: The Perceptron model, linear separability, the XOR failure.	The Reality of Quantitative Backtesting: Discretionary vs mechanical rules.	Implement Perceptron in NumPy; show failure on non-linear XOR truth table.	Script proving XOR non-separability.
Dec 01	Tue	Multi-Layer Perceptrons (MLPs): Hidden layers, activations (Sigmoid, Tanh, ReLU).	Backtesting Pitfall 1: Lookahead Bias; using current close for current entry.	Implement forward pass of 2-layer MLP in pure NumPy with ReLU and Softmax.	Working forward inference script for batch of 32.
Dec 02	Wed	Backpropagation Mathematics: Chain rule derivation for computational graphs.	Backtesting Pitfall 2: Survivorship Bias; testing only current NIFTY 50 members.	Derive $dL/dW1$ and $dL/dW2$ on paper; write gradient updates in NumPy.	Mathematical derivation doc + matching NumPy code.
Dec 03	Thu	Modular neural net from scratch: Layer abstractions, forward and backward.	Backtesting Pitfall 3: Overfitting (Data Snooping); testing 15 indicator parameters.	Assemble object-oriented 3-layer net in NumPy; train it to solve XOR.	Training script showing XOR loss < 0.01.
Dec 04	Fri	Vanishing/exploding gradients; dying ReLU problem; Leaky ReLU and GELU.	Backtesting Pitfall 4: Slippage and Commissions; brokerage and STT friction.	Build 10-layer net with Sigmoid; plot vanishing gradient magnitudes at layer 1.	Notebook visualizing gradient vanishing across layers.
Dec 05	Sat	Lab Build: Train pure NumPy neural network on MNIST digit dataset.	Chart Lab: Add 15 historical trades using strict bar replay (zero lookahead).	Implement batch training in NumPy; hit >92% test accuracy without frameworks.	Framework-free MNIST classifier pushed to repo.
Dec 06	Sun	Review backpropagation partial derivatives and computational graphs.	Recalculate past backtest returns after deducting 0.05% brokerage and STT.	Weekly audit; verify intuitive mastery of backprop chain rule.	Friction-adjusted backtest audit sheet.
Week 10: PyTorch Core, Autograd & 100-Trade Historical Audit					December 07 – December 13, 2026
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Dec 07	Mon	Introduction to PyTorch: Tensors, CPU vs GPU (cuda/mps) device memory allocation.	Quantitative Metrics 1: Maximum Drawdown (MDD) and Drawdown Duration.	Create tensors; benchmark matmul on CPU vs GPU; check device allocation.	Script logging accelerator memory allocation.
Dec 08	Tue	PyTorch Autograd: Computational graphs, requires_grad, .backward(), .grad inspection.	Quantitative Metrics 2: Sharpe Ratio formula, annualized returns, risk-free rate.	Build computational graph; verify manual derivatives with autograd.	Verification script proving autograd equivalence.
Dec 09	Wed	PyTorch torch.nn: Subclassing nn.Module, __init__ and forward, nn.Linear.	Quantitative Metrics 3: Sortino Ratio (downside deviation); Calmar Ratio.	Re-implement MNIST classifier using PyTorch nn.Module; clean inference code.	Clean PyTorch neural network class definition.
Dec 10	Thu	Data handling in PyTorch: Dataset and DataLoader abstractions, batching, workers.	Quantitative Metrics 4: Profit Factor, Win/Loss Ratio, Expectancy scorecard.	Write custom Dataset class loading images; wrap in multi-worker DataLoader.	Working dataset class supporting shuffling & batching.
Dec 11	Fri	Loss Functions and Optimizers: CrossEntropyLoss, BCEWithLogits, SGD, Adam.	Manual Backtesting: Add 15 more trades (85 total); log drawdowns systematically.	Build modular training & validation loop with zeroing, backward, step.	Standardized PyTorch training engine function.

MONTH 3: DAILY STUDY SCHEDULE (WEEKS 11 & 12)

Phase Focus: Deep Regularization, Computer Vision & Paper Trading Deployment

Week 11: Deep Regularization, Optimizers & Paper Trading Setup					December 14 – December 20, 2026
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Dec 14	Mon	Deep Regularization: Dropout mechanics (inverted dropout during training).	Trading Psychology: The emotional toll of drawdown; why rules protect capital.	Implement custom Dropout layer in NumPy; verify activation scaling.	Script verifying inverted dropout numerical behavior.
Dec 15	Tue	Normalization: Batch Normalization (BatchNorm1d, BatchNorm2d) dynamics.	Strategy Drift: The danger of abandoning a verified system during normal drawdowns.	Add BatchNorm1d and Dropout to deep MLP; compare convergence speed.	Comparative loss plots showing faster convergence.
Dec 16	Wed	Weight Initialization: Xavier (Glorot) and He (Kaiming) mathematical proofs.	Capital Allocation: Fixed fractional sizing vs Martingale systems (ruin math).	Initialize network with normal, zeros, and He init; plot layer activations.	Activation histogram plots verifying variance preservation.
Dec 17	Thu	Advanced Optimizers: Momentum, RMSProp, Adam, and AdamW (weight decay).	Paper Trading Setup: Configure TradingView paper trading simulator.	Implement SGD with momentum from scratch in Python; visualize trajectory.	Trajectory visualization overcoming saddle points.
Dec 18	Fri	Experiment Tracking: TensorBoard integration with PyTorch; logging scalars & loss.	Paper Trading Rules: Treat simulated capital with same discipline as real cash.	Instrument training script with SummaryWriter; track train/val loss in real time.	Local TensorBoard dashboard running with loss curves.
Dec 19	Sat	Lab Build: Multi-class neural network; tune with AdamW & LR decay.	Simulation Lab: Execute 3 simulated trades in TradingView paper account.	Implement CosineAnnealingLR scheduler; verify accuracy gains on test set.	Fully tracked PyTorch run with model artifacts.
Dec 20	Sun	Review normalization mathematics and AdamW update equations.	Assess 3 executed paper trades: Did fills match triggers? Log slippage.	Weekly review; confirm zero deviations from trading playbook.	Simulated trade audit sheet logged in journal.
Week 12: Convolutional Vision, Skip Connections & Market Regimes					December 21 – December 27, 2026
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Dec 21	Mon	Computer Vision: Why MLPs fail on high-res spatial data (parameter explosion).	Market Regimes: Trending vs Mean-Reverting vs Volatile/Chop regimes.	Calculate parameter counts of MLP vs Conv layer on 256x256x3 image.	Markdown note proving parameter inefficiency of MLPs.
Dec 22	Tue	Convolutions: Kernels, stride, padding (valid vs same), output shape math.	Strategy Adaptation: Why trend systems fail in chop; identifying ranges.	Implement 2D convolution from scratch in NumPy; apply Sobel edge filter.	Script rendering Sobel horizontal/vertical edge maps.
Dec 23	Wed	Pooling Layers: Max Pooling, Avg Pooling, Global Avg Pooling; translation invariance.	Paper Trading: Scan NIFTY stocks daily; execute triggered setups cleanly.	Build PyTorch CNN with Conv2d, MaxPool2d, Linear layers on CIFAR-10.	Functional CNN architecture passing forward shape tests.
Dec 24	Thu	Modern CNNs: Residual connections (ResNet); degradation problem; skip shortcuts.	Paper Trading: Manage open simulated positions; maintain strict stop orders.	Implement custom ResidualBlock module in PyTorch with identity shortcuts.	Unit-tested ResNet residual block implementation.
Dec 25	Fri	Transfer Learning: Using torchvision.models, layer freezing, classifier heads.	The Myth of Stock Prediction: Why naive LSTM price models overfit and fail.	Load pretrained ResNet-18; freeze backbone; fine-tune classification head.	Transfer learning script with rapid loss convergence.
Dec 26	Sat	Capstone Build: PyTorch Deep	Simulation Lab: Audit	Assemble repo: training	Complete production

MONTH 3: EXTENSION WEEK & MID-PROGRAM CHECKPOINT

Phase Focus: Model Serialization, Quantized Inference & Halfway Milestone Defense

Extension Week: Model Serialization, Quantized Inference & Mid-Program Review December 28, 2026 – January 03, 2027					
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Dec 28	Mon	Model Diagnostics: Confusion matrices, Class Activation Mapping (Grad-CAM).	Non-Stationarity in Finance: Why mean and variance drift across market regimes.	Implement Grad-CAM heatmap visualization over test images.	Grad-CAM visualization overlay script.
Dec 29	Tue	Model Optimization: Quantization basics (FP32 to INT8), TorchScript tracing.	Data Transformations: Stationarity through log returns and fractional diff.	Trace PyTorch model via torch.jit.trace; benchmark inference speed.	Serialized TorchScript model with latency benchmarks.
Dec 30	Wed	PyTorch Best Practices: Seeding, mixed-precision training (torch.cuda.amp).	Financial Feature Engineering: Lagged returns, rolling ATR vs raw price.	Instrument training loop with GradScaler; profile VRAM reduction.	Notebook proving 2x memory reduction via AMP.
Dec 31	Thu	Mid-Program Refactoring: Enforce PEP 8 (ruff/black), write comprehensive READMEs.	Paper Trading Audit: Confirm trade journal matches paper account records.	Run automated linters across all code written during Months 1–3.	Cleaned GitHub repos with zero linting warnings.
Jan 01	Fri	Deep Learning Synthesis: Connect MLPs, CNNs, regularization, and loss surfaces.	The Quantitative Mindset: Trading as statistical experimentation, not opinion.	Write technical synthesis documenting deep learning mechanics learned.	5-page technical markdown synthesis.
Jan 02	Sat	Integration Build: Run full test suite across Months 1–3 codebases.	Simulation Lab: Complete 10 simulated trade updates; record discipline score.	Ensure all environment files (requirements.txt) are pinned and reproducible.	100% test pass rate across verified environments.
Jan 03	Sun	Month 3 Checkpoint: Deep learning exam; explain backprop on paper.	Month 3 Checkpoint: Submit 100-trade backtest audit and 2-week paper journal.	Score mid-program milestone; celebrate completing first half of roadmap.	Month 3 milestone signed off.

Halfway Program Audit & Performance Review			Months 1–3 Milestones Completed
DOMAIN	CORE CAPABILITIES DEMONSTRATED	VERIFIED DELIVERABLES IN PORTFOLIO	
Mathematical & Classical ML	Vectorized array math, linear algebra, loss derivation, Scikit-learn pipelines, Optuna tuning.	Automated EDA Engine + Enterprise Default Predictor.	
Deep Learning & Vision	Computational graphs, manual backprop in NumPy, PyTorch Module/DataLoader, CNNs, ResNets.	Scratch MNIST Classifier + PyTorch Vision Pipeline.	
Systematic Trading Track	Market structure, S/R, dynamic ATR stops, 1% risk sizing, transaction cost friction modeling.	100-Trade Empirical Audit + Verified Paper Journal.	

MONTH 4: DAILY STUDY SCHEDULE (WEEKS 13 & 14)

Phase Focus: Sequence Models, Self-Attention & Vectorized Quantitative Backtesting

Week 13: Sequence Modeling, RNNs & Financial Time Series Analysis					January 04 – January 10, 2027
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Jan 04	Mon	Sequence Modeling: Recurrent Neural Networks (RNNs), hidden state propagation.	Quant Finance in Python: yfinance setup, multi-year daily stock ingestion.	Code unrolled vanilla RNN cell in NumPy; trace hidden state transitions.	Script executing manual recurrent sequence passes.
Jan 05	Tue	Gradients in RNNs: Long Short-Term Memory (LSTM) gates (forget, input, output).	Time Series in Pandas: Daily returns, log returns, rolling volatility.	Implement LSTM cell forward pass in PyTorch; verify tensor shapes (B, T, D).	Working PyTorch LSTM sequence processing script.
Jan 06	Wed	Gated Recurrent Units (GRUs): Simplified gating; parameter counts vs LSTMs.	Stationarity: Augmented Dickey-Fuller (ADF) test on prices vs log returns.	Train PyTorch GRU on synthetic sine wave; predict autoregressively.	Plot showing multi-step forward sequence forecast.
Jan 07	Thu	Seq2Seq Architectures: Encoder-decoder structures; sequence bottleneck.	Vectorized Backtesting: Shifting series in Pandas; executing on open[t+1].	Build encoder-decoder architecture in PyTorch for sequence mapping.	Functional Seq2Seq PyTorch model compiling cleanly.
Jan 08	Fri	Tokenization: Character vs Word vs Subword; Byte-Pair Encoding (BPE) algorithm.	Vectorized EMA Strategy: Generating signals; avoiding lookahead via shifting.	Implement basic character-level tokenizer with encode() and decode().	Reusable tokenizer mapping text to integer arrays.
Jan 09	Sat	Lab Build: Train autoregressive character-level LSTM on Shakespeare text.	Quant Lab: Vectorized backtest for 20/50 EMA crossover on NIFTY (2018–2026).	Sample generated text with temperature scaling; evaluate output.	Text generator script + clean vector backtest script.
Jan 10	Sun	Review recurrent gating equations and vectorized Pandas backtesting mechanics.	Inspect equity curve of EMA strategy; calculate Sharpe and Max Drawdown.	Weekly review; verify zero lookahead in Pandas signal shifting.	Vectorized strategy performance report.
Week 14: Attention Mechanisms & Building Transformers from Scratch					January 11 – January 17, 2027
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Jan 11	Mon	The Breakdown of Recurrence: Why Transformers replaced RNNs (parallelization).	Vectorized Backtesting 2: Adding transaction costs and slippage to Pandas model.	Write technical brief comparing RNN steps vs Transformer parallel scaling.	Markdown analysis of recurrent computational bottlenecks.
Jan 12	Tue	Attention Foundations: Query, Key, Value intuition; soft dictionary lookups.	Benchmark Comparison: Compare strategy equity curve against Buy-and-Hold NIFTY.	Code Scaled Dot-Product Attention from scratch in pure NumPy.	Script matching PyTorch scaled_dot_product_attention.
Jan 13	Wed	Multi-Head Attention (MHA): Splitting embedding dimensions; linear projections.	Volatility Targeting: Sizing historical positions dynamically using 20-day ATR.	Implement multi-head attention module in PyTorch using nn.Linear layers.	Unit-tested PyTorch module executing MHA.
Jan 14	Thu	Positional Encodings: Permutation invariance; sinusoidal encoding formulation.	Parameter Optimization: Grid search over EMA parameters; curve-fitting hazards.	Code sinusoidal positional encoding in PyTorch; visualize frequency heatmaps.	Heatmap plot displaying unique positional signatures.
Jan 15	Fri	Transformer Architecture: Pre-LayerNorm vs Post-LayerNorm; Feed-Forward Networks.	Walk-Forward Analysis: In-sample optimization vs out-of-sample testing windows.	Assemble complete Decoder-only Transformer block in PyTorch.	Transformer block class passing automated assertions.
Jan 16	Sat	Lab Build: Assemble Mini-Transformer Language Engine	Quant Lab: Code walk-forward optimization	Train mini-transformer on small text corpus;	Decoder-only Transformer training

MONTH 4: DAILY STUDY SCHEDULE (WEEKS 15 & 16)

Phase Focus: Hugging Face Ecosystem, Vector Embeddings & Quantitative Screening

Week 15: Hugging Face Ecosystem & Vectorized Pandas Backtesting					January 18 – January 24, 2027
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Jan 18	Mon	Hugging Face: transformers library, AutoModel, AutoTokenizer, pipeline abstractions.	Quant Filters: Incorporating RSI and volume thresholds into equity backtester.	Load DistilBERT via Hugging Face; run sentiment classification on headlines.	Script returning sentiment scores for stock news.
Jan 19	Tue	Fine-tuning Encoders: Data preparation via HF datasets; input sequence formatting.	Strategy Execution Drift: Comparing backtest fills against simulated paper fills.	Fine-tune DistilBERT for sentiment classification via HF Trainer API.	Working training script outputting model checkpoints.
Jan 20	Wed	Modern LLMs: Decoder-only families (Llama, GPT); autoregressive pre-training.	Paper Trading: Execute simulated setups; log psychological urges to deviate.	Write generation script with Greedy, Top-k, and Top-p (Nucleus) sampling.	Script illustrating sampling parameter impact on outputs.
Jan 21	Thu	Parameter-Efficient Fine-Tuning (PEFT): LoRA and QLoRA low-rank adaptation math.	Risk Audit: Verify simulated portfolio drawdown remains within predefined limit.	Instrument model with PEFT/LoRA; inspect trainable parameter count (<1%).	LoRA fine-tuning script with >99% frozen weights.
Jan 22	Fri	Vector Embeddings: Dense vector representations, cosine distance, sentence-transformers.	Quant Screening: Python script to screen NSE stocks with ROCE > 15% and P/E < 25.	Embed 100 sentences via sentence-transformers; plot similarity matrix.	Notebook visualizing semantic clustering of sentences.
Jan 23	Sat	Lab Build: Assemble Vectorized Quant Research Engine in Python.	Simulation Lab: Audit paper positions; compute rolling simulated Sharpe ratio.	Build modular backtest framework evaluating multi-asset rule sets.	GitHub repository with quant_engine.py module.
Jan 24	Sun	Review LoRA low-rank updates and sentence embedding geometry.	Scan Indian market breadth: % of NIFTY stocks trading above their 50-day EMA.	Weekly retrospective; record quantitative screening results.	Market breadth indicator report logged.
Week 16: Embeddings, Dense Retrieval & Quantitative Strategy Verification					January 25 – January 31, 2027
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Jan 25	Mon	Vector Databases: The need for indexing; exact vs Approximate Nearest Neighbors (ANN).	Quantitative Metrics: Tracking max daily loss, profit factor, recovery factor.	Implement flat vector search index from scratch in NumPy using cosine metric.	Script running top-k search over 10,000 vectors.
Jan 26	Tue	ANN Indexing: Hierarchical Navigable Small World (HNSW) and Inverted File (IVF).	Strategy Robustness: Vary indicator parameters by +/- 10% to check fragility.	Initialize local ChromaDB/FAISS index; insert 1,000 docs; benchmark query speed.	Benchmark script showing sub-millisecond retrieval.
Jan 27	Wed	Dense Retrieval: Chunking strategies (fixed, recursive, semantic) and overlap.	Paper Trading: Execute setups cleanly; record discipline rating for each fill.	Write custom chunking module; evaluate impact on semantic preservation.	Script comparing fixed vs recursive document chunking.
Jan 28	Thu	Re-ranking: Cross-encoders vs Bi-encoders; re-ranking top-k vector candidates.	Algo Trading Microstructure: API architecture of Indian brokers (Zerodha Kite).	Build two-stage pipeline: Dense retrieval followed by cross-encoder ranker.	Pipeline demonstrating improved search precision.
Jan 29	Fri	LLM Context Windows: Lost-in-the-middle problem, prompt formatting for RAG context.	SEBI Regulatory Framework: Rules regarding retail automated trading in India.	Construct evaluation script measuring accuracy over needle-in-haystack docs.	Markdown analysis of context window limits.

MONTH 5: DAILY STUDY SCHEDULE (WEEKS 17 & 18)

Phase Focus: Production RAG, Hybrid Search & Derivatives Market Mechanics

Week 17: Production RAG Architectures & Futures Market Mechanics					February 01 – February 07, 2027
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Feb 01	Mon	RAG Architecture: Ingestion, indexing, retrieval, synthesis pipeline design.	Introduction to Derivatives: Risk transfer, hedging vs speculation, forward contracts.	Sketch complete system architecture diagram for enterprise RAG.	Architectural system blueprint in repo.
Feb 02	Tue	Naive RAG vs Advanced RAG: Limitations of naive retrieval (hallucinations, noise).	Futures Contracts: Specifications, lot sizes, tick sizes, MTM settlement on NSE.	Build working naive RAG script using LangChain; query technical docs.	Working baseline RAG query script.
Feb 03	Wed	Document Parsing: Extracting text, tables, and metadata from structured PDFs.	Futures Pricing: Spot-Futures parity, cost of carry, Contango vs Backwardation.	Ingest corporate annual report PDF; extract financial tables to Markdown.	Script parsing complex PDF tables cleanly.
Feb 04	Thu	Query Transformation: Query rewriting, sub-query decomposition, HyDE embeddings.	Futures Margins: Initial, maintenance, Span, and exposure margins on NSE.	Implement query decomposition breaking complex prompt into 3 sub-queries.	Query decomposition and multi-retrieval module.
Feb 05	Fri	Hybrid Search: Combining BM25 keyword search with dense vectors via RRF.	The Leverage Trap: Calculating ruin probability of trading futures undercapitalized.	Code Reciprocal Rank Fusion (RRF) algorithm merging BM25 and ChromaDB.	Script proving hybrid retrieval superior keyword recall.
Feb 06	Sat	Lab Build: Hybrid RAG system parsing earnings releases; evaluate recall.	Derivatives Lab: Python futures pricing calculator; compute basis and carry cost.	Run 20 domain queries through Hybrid RAG; benchmark response accuracy.	Hybrid RAG app with benchmark suite pushed to repo.
Feb 07	Sun	Review Reciprocal Rank Fusion mathematics and futures margin calculations.	Inspect NIFTY futures chain on NSE; calculate basis (Futures minus Spot).	Weekly review; document why leverage ruins retail traders.	Futures basis calculation sheet logged.
Week 18: Advanced RAG, Hybrid Search & Options Fundamentals					February 08 – February 14, 2027
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Feb 08	Mon	Context Management: Window pruning, dynamic compression, citation attribution.	Options Basics: Calls and Puts; buyer rights vs seller obligations.	Write context compressor removing low-similarity sentences before generation.	Script reducing token count while retaining core facts.
Feb 09	Tue	RAG Evaluation: Ragas metrics (Faithfulness, Answer Relevance, Context Precision).	Moneyness: In-the-Money (ITM), At-the-Money (ATM), Out-of-the-Money (OTM).	Run automated Ragas evaluation across synthetic question-context triples.	Automated test script outputting Faithfulness scores.
Feb 10	Wed	Prompt Engineering: Structured JSON outputs, Pydantic schema validation.	Intrinsic vs Extrinsic (Time) Value: Why OTM options have 100% time value.	Build Pydantic schema validating structured financial metrics from LLM.	Type-safe JSON extraction script.
Feb 11	Thu	Local LLM Inference: Serving open models via llama.cpp / Ollama; GGUF formats.	Option Payoffs: Hockey-stick payoff diagrams for Long Calls and Long Puts.	Configure local Ollama serving; benchmark tokens-per-second on local GPU.	Script executing local LLM inference without APIs.
Feb 12	Fri	LLM Caching: Semantic caching (GPTCache), exact key-value caching; cost control.	Option Selling Mechanics: Asymmetry of limited profit vs undefined risk.	Implement local semantic cache storing prompt embeddings for fast hits.	Script showing <10ms cached semantic query returns.

MONTH 5: DAILY STUDY SCHEDULE (WEEKS 19 & 20)

Phase Focus: Tool-Augmented Agents, Options Greeks & Containerized Serving

Week 19: Tool-Augmented Agents & Options Greeks Mechanics

February 15 – February 21, 2027

DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Feb 15	Mon	Agentic AI: The ReAct framework (Reasoning + Acting); Thought-Action-Observation.	The Greeks 1: Delta - directional sensitivity, hedge ratio, probability proxy.	Code basic ReAct loop in Python invoking calculator and search tools.	Working autonomous multi-step reasoning script.
Feb 16	Tue	Tool Calling: Function calling schemas, JSON schema definitions, arg parsing.	The Greeks 2: Theta - time decay; exponential acceleration in final 30 days.	Define custom tool fetching stock price; bind to LLM; execute tool calls.	Script executing tool calls from natural text prompts.
Feb 17	Wed	Multi-Agent Systems: Specialized agents, supervisor/router architectures.	The Greeks 3: Gamma - rate of change of Delta; gamma risk near expiry.	Build two-agent pipeline: Researcher gathers data, Writer drafts report.	Working collaborative multi-agent execution script.
Feb 18	Thu	Agent Memory: Short-term conversational buffer vs long-term vector store memory.	The Greeks 4: Vega - sensitivity to Implied Volatility; earnings IV crush.	Implement state machine tracking user goals and completed agent sub-tasks.	Stateful agent execution script across multiple turns.
Feb 19	Fri	Guardrails in Agents: Infinite loop prevention, execution budgets, input sanity.	IV vs Historical Volatility (HV): The volatility smile and skew on NIFTY index.	Implement hard execution guards: Max loop counters, timeouts, rate limits.	Production-safe agent wrapper preventing budget runaway.
Feb 20	Sat	Lab Build: Autonomous financial agent fetching prices & drafting analysis.	Derivatives Lab: Python Black-Scholes Greeks calculator; plot Theta decay.	Equip agent with price fetcher, PE calculator, and report generator.	Financial analysis agent script outputting dossiers.
Feb 21	Sun	Review ReAct execution traces and Black-Scholes formula intuition.	Inspect options chain of earnings stock; record ATM IV before earnings.	Weekly review; synthesize options Greeks and agent safety loops.	Implied volatility case study document.

Week 20: Serving, Containerization (Docker) & Derivatives Risk Auditing

February 22 – February 28, 2027

DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Feb 22	Mon	Production Serving: FastAPI framework, async endpoints, Pydantic schemas.	Options Realities: Why buying far OTM options has negative expected value.	Build FastAPI service serving model inference with schema validation.	Functional REST API accepting POST predictions.
Feb 23	Tue	High-Performance Serving: StreamingResponse, server-sent events for LLMs.	Defined-Risk Spreads: Bull Call Spreads and Bear Put Spreads (hedging).	Implement token streaming on FastAPI inference endpoint.	Working client script streaming tokens in real time.
Feb 24	Wed	Docker Fundamentals: Dockerfile syntax, base images, containerizing Python apps.	Hedging Mechanics: Using index puts or futures to hedge equity portfolios.	Write multi-stage Dockerfile containerizing FastAPI service cleanly.	Optimized Docker image built with size < 500 MB.
Feb 25	Thu	Docker Compose: Multi-service stacks (FastAPI, ChromaDB, Redis cache).	Why F&O Destroys Beginners: SEBI study (93% of retail traders lose in F&O).	Write docker-compose.yml running API backend and persistent ChromaDB.	Multi-container stack launching via docker-compose up.
Feb 26	Fri	Continuous Integration (CI): GitHub Actions; automated pytest and linting.	Paper Trading Preparation: Consolidate complete playbook for Month 6 live sim.	Create GitHub Actions workflow running tests on every repository push.	Active GitHub repo with green passing CI badges.
Feb 27	Sat	Lab Build: Deploy	Simulation Prep: Set up	Write integration tests	100% passing

MONTH 6: DAILY STUDY SCHEDULE (WEEKS 21 & 22)

Phase Focus: Capstone Engineering (FinSpectra Platform) & Live Forward Paper Trading

Week 21: Capstone Ingestion Pipeline, Architecture & Live Simulation					March 01 – March 07, 2027
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Mar 01	Mon	Capstone Architecture: FinSpectra Platform design, components, data flows.	Forward Paper Trading Day 1: Pre-market routine; check NIFTY trend.	Write complete technical specification document for FinSpectra.	Technical specification doc in capstone repo.
Mar 02	Tue	Data Ingestion: Ingest Indian corporate earnings transcripts and NSE daily OHLCV.	Forward Paper Trading Day 2: Monitor market action; log triggered setups.	Build automated scrapers pulling transcript text and price series.	Ingestion pipeline with retry error handling.
Mar 03	Wed	Vector Store Construction: Chunk transcripts; embed and index in ChromaDB.	Forward Paper Trading Day 3: Execute simulated orders; enforce ATR stops.	Index 50 earnings transcripts with metadata (ticker, quarter, year).	Verified persistent vector index of transcripts.
Mar 04	Thu	Hybrid Retrieval: Connect BM25 and vector search; add cross-encoder ranker.	Forward Paper Trading Day 4: Post-market review; log intraday chop.	Implement hybrid search returning top-5 financial passages with citations.	Python module returning ranked document chunks.
Mar 05	Fri	Financial Extraction: Prompt pipelines extracting guidance, revenue risks.	Forward Paper Trading Day 5: Weekly reconciliation; calculate paper R.	Build extraction pipeline parsing structured guidance from transcripts.	Validated Pydantic models populated with data.
Mar 06	Sat	Capstone Build: Integrate ingestion, vector indexing, and extraction.	Simulation Lab: Audit Week 1 paper trades; verify strict rule compliance.	Write end-to-end integration tests for backend ingestion service.	Integrated backend core library pushed to repo.
Mar 07	Sun	Review Capstone backend pipeline throughput and latency bottlenecks.	Scan NIFTY 100 stocks; build candidate watchlist for Week 22 simulation.	Weekly retrospective; ensure zero unhandled backend exceptions.	Updated watchlist of 5 candidates logged.
Week 22: Analytics Engine, ML Modeling & Systematic Execution					March 08 – March 14, 2027
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Mar 08	Mon	Quantitative Engine: Build analytics library for returns, volatility, drawdowns.	Forward Paper Trading Day 6: Scan candidate setups; execute simulated entry.	Write pure Pandas analytics module computing 15 financial metrics.	Self-contained quant analytics module with tests.
Mar 09	Tue	Sentiment Classifier: Integrate fine-tuned transformer scoring transcript tone.	Forward Paper Trading Day 7: Manage open positions; trail stop orders.	Connect sentiment model to extraction pipeline; score quarterly tone.	Pipeline generating sentiment scorecards.
Mar 10	Wed	Regime Classifier: Implement Random Forest model classifying volatility states.	Forward Paper Trading Day 8: Maintain journal; log emotional impulses.	Train regime model categorizing historical market windows into chop/trend.	Model classifying market regime from daily prices.
Mar 11	Thu	FastAPI Integration: Wrap analytics, RAG, and models in asynchronous API.	Forward Paper Trading Day 9: Execute simulated orders; log fill slippage.	Build REST endpoints: /query-rag, /financial-metrics, /stock-summary.	Functional REST API exposing all backend capabilities.
Mar 12	Fri	API Optimization: Implement response streaming and Redis caching for queries.	Forward Paper Trading Day 10: Weekly reconciliation; calculate win rate.	Add streaming endpoint /stream-rag; benchmark latency with Redis cache.	Latency benchmark showing <50ms cached responses.
Mar 13	Sat	Capstone Build: Interactive frontend using Streamlit/Next.js for	Simulation Lab: Audit simulated trades against journal; check execution	Connect UI to FastAPI backend; build dashboards for queries	Interactive web app running and communicating with

MONTH 6: DAILY STUDY SCHEDULE (WEEKS 23 & 24)

Phase Focus: Multi-Agent Orchestration, Cloud Deployment & Paper Trade Audit

Week 23: Production UI, Multi-Agent Logic & Slippage Tracking					March 15 – March 21, 2027
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Mar 15	Mon	Multi-Agent Orchestrator: Route query to RAG, Quant Tool, or Sentiment Engine.	Forward Paper Trading Day 11: Execute daily routine; take zero unverified setups.	Build router agent directing user requests to appropriate specialized tool.	Intelligent query router module passing routing tests.
Mar 16	Tue	Agent Tool Integration: Equip agent with NSE price fetcher and technical indicator tools.	Forward Paper Trading Day 12: Manage simulated trades; log trailing stops.	Test agent tool calling across complex multi-step user prompts.	Trace log showing agent invoking multiple tools.
Mar 17	Wed	Evaluation Suite: Build test harness measuring retrieval and synthesis accuracy.	Forward Paper Trading Day 13: Execute market scans; log false breakout patterns.	Run evaluation script across 50 benchmark queries; calculate faithfulness.	Benchmark report confirming >85% faithfulness.
Mar 18	Thu	Robustness & Fallbacks: Graceful handling of missing filings and network timeouts.	Forward Paper Trading Day 14: Review mindset; avoid emotional attachment to sim.	Stress-test platform with malformed inputs and missing document chunks.	Error-handling layer returning clean user messages.
Mar 19	Fri	Documentation & Packaging: Write API docs, user guides, Mermaid diagrams.	Forward Paper Trading Day 15: Weekly reconciliation; compute cumulative return.	Write professional GitHub README with architecture diagrams and quickstart.	Production README ready for recruiter review.
Mar 20	Sat	Capstone Build: Multi-service containerization via docker-compose.	Simulation Lab: Audit 15 paper trades; verify 100% adherence to risk limits.	Launch entire stack (UI, FastAPI, ChromaDB) with single command.	Verified docker-compose deployment working locally.
Mar 21	Sun	Review container resource utilization and backend latency profiles.	Scan global markets: US indices, Dollar Index (DXY), crude oil impact on India.	Weekly retrospective; ensure zero Docker configuration warnings.	Global macro correlation notes logged.
Week 24: Cloud Deployment, Monitoring & Final 6-Month Defense					March 22 – March 28, 2027
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Mar 22	Mon	Cloud Deployment Prep: Select hosting targets; optimize container footprints.	Forward Paper Trading Day 16: Maintain discipline; verify entry checklist.	Optimize Docker image size using multi-stage builds and minimal base.	Lightweight container image ready for deployment.
Mar 23	Tue	Production Cloud Deployment: Deploy FinSpectra platform; configure SSL and secrets.	Forward Paper Trading Day 17: Log paper fills; calculate execution latency.	Deploy backend and frontend to cloud hosting; configure domain and HTTPS.	Live, publicly accessible web URL running platform.
Mar 24	Wed	Observability & Monitoring: Implement structured logging and error tracking.	Forward Paper Trading Day 18: Manage exits; ensure stop orders triggered on time.	Instrument FastAPI with structured JSON logging and error monitoring.	Live monitoring dashboard displaying API health.
Mar 25	Thu	Portfolio Showcase: Create 3-minute video showing architecture and live demo.	Forward Paper Trading Day 19: Execute daily scans; record chart observations.	Record professional technical walkthrough of FinSpectra for GitHub.	HD product demo video embedded in repo README.
Mar 26	Fri	Codebase Final Polish: Enforce strict mypy type hints, docstring coverage.	Forward Paper Trading Day 20: Conclude 20-day forward paper campaign.	Run final code quality audit across all modules; fix any code smells.	Zero-defect codebase adhering strictly to PEP 8.
Mar 27	Sat	Lab Dry Run: Execute full end-	Simulation Lab: Compile	Verify zero memory leaks	Polished live

MONTH 6: FINAL WRAP-UP & GRADUATION MILESTONE

Phase Focus: Interview Readiness, Personal Trading Constitution & Career Portfolio

Final Program Wrap-Up: Career Portfolio Polish, Interview Prep & System Defense					March 29 – April 04, 2027
DATE	DAY	AI/ML SESSION (19:00–22:00)	TRADING TRACK (22:10–23:00)	PRACTICAL TASK	CONCRETE OUTPUT
Mar 29	Mon	Portfolio Presentation: Clean up GitHub profile, pin key repos, write case study.	Trading Journal Archiving: Archive paper trades into structured Notion database.	Write technical article detailing system architecture of FinSpectra.	Published technical case study walking through design.
Mar 30	Tue	Interview Prep 1: Review ML theory (bias-variance, loss functions, backprop).	Trading Constitution: Write binding Personal Trading Constitution & limits.	Conduct mock interview on ML fundamentals; record clear answers.	Doc containing answers to top 25 ML questions.
Mar 31	Wed	Interview Prep 2: Review deep learning & LLM system design (scaling laws, RAG).	Financial Evaluation: Assess psychological readiness for eventual live trading.	Diagram complete architecture for 3 common AI system design problems.	System design diagrams and architectural notes.
Apr 01	Thu	GitHub Repo Audit: Verify all links, tests, docker files work in fresh clone.	Risk Contract: Establish rule: Max ₹10,000 capital if ever trading live in future.	Execute clone-and-run test in fresh VM; verify single-command installation.	Documented test verifying 100% reproducible install.
Apr 02	Fri	Resume & Career Assets: Update resume with quantitative project bullet points.	Strategy Review: Final audit of mechanical swing system: Win rate & expectancy.	Update professional resume highlighting AI/ML deliverables & metrics.	Clean, ATS-friendly technical resume ready to send.
Apr 03	Sat	Capstone Defense: Present FinSpectra in mock engineering defense session.	Trading Defense: Calculate exact position sizing for 5 setups in under 30s.	Deliver 20-minute technical presentation covering trade-offs and latency.	Recorded technical presentation of capstone.
Apr 04	Sun	Program Graduation: Final review; establish lifelong study schedule.	Program Graduation: Seal trading journal; maintain weekly market habits.	Review full progress from Oct 2026 to Apr 2027; set career goals.	Program Complete: Verified portfolio deployed.

Personal Trading Constitution & Final Competency Checklist			Graduation Sign-Off
CONTRACT CLAUSE	BINDING PERSONAL RULE	ENFORCEMENT & VERIFICATION	
Max Risk Per Trade	Never risk more than 1.0% of total capital on any single trading idea.	Pre-trade position calculator verified.	
Mandatory Invalidation Stop	Every entry must have a technical ATR-based stop order entered immediately.	Zero mental stops permitted.	
Anti-Gambling Directive	Zero telegram tips, zero unhedged options buying, zero intraday revenge trading.	Permanent ban on discretionary trades.	
Primary Career Primacy	AI/ML software engineering remains primary career. Trading remains disciplined hobby.	Academic & job commitments take 100% priority.	